

WE ARE STUDYING AT 4 WHY DON'T YOU DO

WORKED OUT MEMO
TAKE IT AT YOUR OWN RISK
if The ANSWERS A WRONG DON'T
come TO ME A TELL ME SHANT

- (a) Gee 'n uitdrukking vir die kinetiese wrywingskrag F_k in terme van die massa m van 'n voorwerp.
Give an expression for the kinetic friction force F_k in terms of the mass m of an object.

$$F_k = \mu_k N$$

- (b) 'n Voorwerp (massa 20 kg) kan deur 'n baie sterk wind oor 'n woestyn gesleep word. Indien die kinetiese wrywingskoeffisiënt 0,8 is, bereken die horisontale krag wat nodig is om die voorwerp oor die woestynsand te sleep teen 'n konstante speed.
An object (mass 20 kg) can be dragged over the desert floor by a very strong wind. If the coefficient of kinetic friction is 0,8, what horizontal force is needed to maintain the stone's motion at a constant speed.

= 8 D



$$\Sigma F_y = 0 \quad N - W = 0$$

$$F_k - F_A = 0$$

$$W = N = 196 \text{ N}$$

$$F_A = F_k$$

$$F_k = \mu_k N$$

$$F_A = 157 \text{ N}$$

$$= (0,8)(196)$$

$$= 156,8 \approx 157 \text{ N}$$

- (c) Die sleurkrag op die voorwerp is $D = \frac{1}{2} C_p A v^2$. Bereken die windsnelheid (in km/h) op die grond indien die sleurkoeffisiënt 0,8 en die digtheid van lug $1,21 \text{ kg/m}^3$ is. Die deursnit-oppervlakte van die voorwerp is $0,04 \text{ m}^2$.
The drag force on the object is $D = \frac{1}{2} C_p A v^2$. Determine the wind speed (in km/h) along the ground if the drag coefficient is 0,8 and the density of air is $1,21 \text{ kg/m}^3$. The cross-sectional area of the object is $0,04 \text{ m}^2$.

$$D = \frac{1}{2} C_p A v^2$$

$$\therefore v = 90,1$$

$$\Rightarrow v = \sqrt{\frac{2D}{C_p A}}$$

$$3,6$$

$$v = 25,0 \text{ km h}^{-1}$$

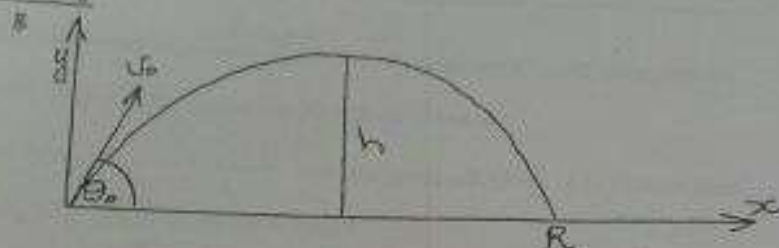
$$= \sqrt{\frac{2(157)}{(0,8)(1,21)(0,04)}}$$

$$= 90,15^{-1}$$

VRAAG/QUESTION 2

- (a) 'n Projektiel word oor 'n horisontale oppervlak gelanseer teen 'n hoek θ_0 met die horisontaal. Die beginsnelheid is v_0 en die maksimum trefafstand is R . Toon aan dat die vlugtyd gegee word deur $T = \frac{2 v_0 \sin \theta_0}{g}$.

A projectile is launched over a horizontal plane at an angle θ_0 relative to the horizontal. The initial velocity is v_0 and the range is R . Show that the time of flight is given by $T = \frac{2 v_0 \sin \theta_0}{g}$.



Time at Max. height

$$v_{yf} = v_{yi} + a_y T$$

$$0 = v_0 \sin \theta_0 - g T$$

$$g T = v_0 \sin \theta_0$$

$$T = \frac{v_0 \sin \theta_0}{g}$$

Total Time

$$T_1 + T_1 = \text{Total Time}$$

$$\frac{v_0 \sin \theta_0}{g} + \frac{v_0 \sin \theta_0}{g}$$

$$T = \frac{2 v_0 \sin \theta_0}{g}$$

(4)

- (b) Toon aan dat die maksimum hoogte wat bereik word gegee word deur $h = \frac{v_0^2 \sin^2 \theta_0}{2g}$.

Show that the maximum height is given by $h = \frac{v_0^2 \sin^2 \theta_0}{2g}$.

$$h = v_{yi} T - \frac{1}{2} a_y T^2$$

$$h = (v_0 \sin \theta_0) \left(\frac{v_0 \sin \theta_0}{g} \right) - \frac{1}{2} g \left(\frac{v_0 \sin \theta_0}{g} \right)^2$$

$$h = \frac{v_0^2 \sin^2 \theta_0}{g} - \frac{v_0^2 \sin^2 \theta_0}{2g} \Rightarrow \frac{v_0^2 \sin^2 \theta_0}{2g}$$

(4)

Vraag / Question 1(c)(Vervolg)(Continue) /

- (c) Met $h = \frac{R}{3}$, toon aan dat $v_0 \sin \theta_0 = \sqrt{\frac{Rg}{3}}$
 With $h = \frac{R}{3}$, show that $v_0 \sin \theta_0 = \sqrt{\frac{Rg}{3}}$.

$$\frac{R}{6} = \frac{v_0^2 \sin \theta_0}{2g} \Rightarrow v_0 \sin \theta_0 = \sqrt{\frac{3R}{3}} \quad (2)$$

- (d) Met die reikwydte $R = \frac{v_0^2 \sin 2\theta_0}{g}$ en die gegewens in (c), toon aan dat

$$v_0 \cos \theta_0 = \frac{1}{2} \sqrt{3gR} \quad \text{indien } \sin 2\theta = 2 \sin \theta \cos \theta.$$

With the range $R = \frac{v_0^2 \sin 2\theta_0}{g}$ and the formula given in (c), show that

$$v_0 \cos \theta_0 = \frac{1}{2} \sqrt{3gR} \quad \text{if } \sin 2\theta = 2 \sin \theta \cos \theta.$$

$$R = \frac{v_0^2 \sin 2\theta_0}{g}$$

$$\Rightarrow \frac{\sqrt{3}}{2} \left[\sqrt{Rg} \right]$$

$$Rg = 2v_0 \cdot v_0 \sin \theta_0 \cos \theta_0$$

$$Rg = 2v_0 \cdot \frac{1}{2} \sqrt{3gR} \cdot \cos \theta_0$$

$$\Rightarrow v_0 \cos \theta_0 = \frac{1}{2} \left[\frac{Rg}{\frac{1}{2} \sqrt{3gR}} \right] \Rightarrow \frac{1}{2} \sqrt{3Rg}$$

$$\Rightarrow \frac{\sqrt{3}}{2} \left[(Rg)^{\frac{1}{2}} \right]$$

- (e) Toon aan dat die tyd wat nodig is om die piekhoogte te bereik gegee word deur $t_p = \sqrt{\frac{R}{3g}}$. (3)

Show that the time it takes to reach the top is given by $t_p = \sqrt{\frac{R}{3g}}$.

Time at maximum height

$$t_p = \frac{v_0 \sin \theta_0}{g}$$

$$= \frac{\frac{1}{2} \sqrt{3gR}}{g} = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{g}} \sqrt{R} = \sqrt{\frac{R}{3g}} \quad (3)$$

- (f) Indien $R = 19,6$ m, bepaal die vliegtyd van die projektiel.
 If $R = 19,6$ m, determine the time of flight of the projectile.

$$t_p = \sqrt{\frac{R}{3g}} = \sqrt{\frac{19,6}{3(9,8)}} \quad \text{total time} = 2 t_p$$

$$\Rightarrow 0,8163 \quad \text{total time} = 1,6326$$

- (g) Toon aan dat die spoed van die projektiel by die toppunt van die baan gegee word deur $v_{top} = \frac{1}{2}\sqrt{3gR}$. (3)

Show that the speed of the projectile at the top of the trajectory is given by $v_{top} = \frac{1}{2}\sqrt{3gR}$.

$$v_f = \sqrt{v_{xf}^2 + v_{yf}^2} \Rightarrow v_{yf} = \sqrt{\left(\frac{1}{2}(\sqrt{3gR})\right)^2}$$

$$0 = \sqrt{\left(\frac{1}{2}(\sqrt{3gR})\right)^2 - v_{yf}^2} = \frac{1}{2}\sqrt{3gR} \quad (2)$$

- (h) Toon aan dat die hoek waarteen die projektiel gelanseer word is $33,7^\circ$.
 Show that the angle at which the projectile has been launched is $33,7^\circ$.

$$\frac{v_0 \sin \theta_0}{v_0 \cos \theta_0} = \frac{\sqrt{\frac{1}{3}gR}}{\frac{1}{2}\sqrt{3gR}} \Rightarrow \tan^2 \theta = \frac{4}{9} \quad \theta = \tan^{-1}\left(\frac{2}{3}\right)$$

$$\tan \theta = \frac{2}{3} \quad \theta = 33,69^\circ \quad (4)$$

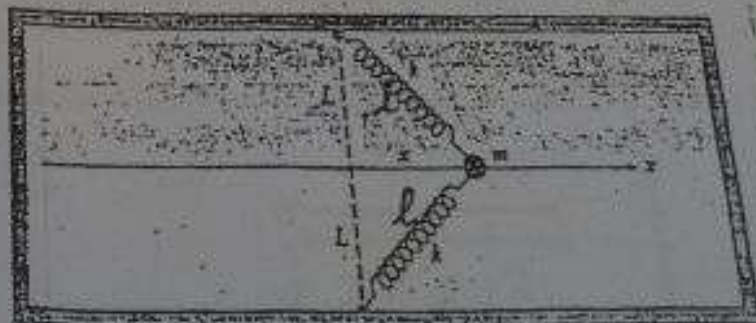
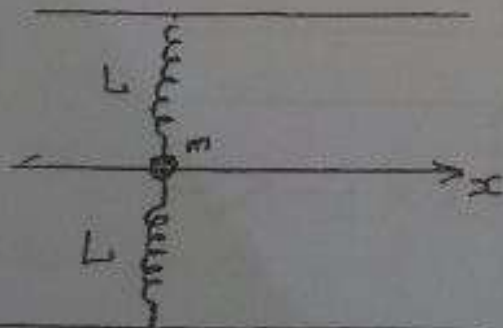
VRAAG / QUESTION 3

- (a) 'n Veer met veerkonstante k word 'n afstand x uitgerek. Skryf 'n uitdrukking neer vir die krag F_s in die veer.
 A spring with spring constant k is stretched over a distance x . Give an expression for the spring force F_s .

$$F_s = -kx$$

(1)

- (b) 'n Massa is vas aan twee identiese vere, beide met dieselfde veerkonstante k . Die massa word 'n afstand x langs die x -as getrek vanaf die oorspronklike onuitgerekte posisie.
 A mass is attached between two identical springs. The spring constant of each spring is k . The mass is pulled over a distance x along the x -axis from the initially unstretched position.



- (i) Skryf 'n uitdrukking neer vir die (nuwe) uitgerekte lengte l van die veer.
Give an expression for the (new) stretched length l of the spring.

$$\sqrt{L^2 + x^2}$$

(1)

- (ii) Met hoeveel het elke veer verleng vanaf sy ongespanne oorspronklike onuitgerekte lengte L ?
Give the elongation of each spring (relative to the unstretched length L).

$$\sqrt{L^2 + x^2} - L$$

(1)

- (iii) Skryf 'n uitdrukking neer vir die grootte van die krag F_s in elke veer.
Give an expression for the magnitude of the force F_s in each spring.

$$F_s = k(\sqrt{x^2 + L^2} - L)$$

(2)

- (iv) Die y-komponente van die kragte in die twee veer word gegee deur:
The y-components of the forces in the two springs are given by:

$$\sum (F_s)_y = 0$$

(2)

- (v) - Toon aan dat die som van die kragte van die twee veer in die x-as rigting gegee word

$$\text{deur } \vec{F} = -2kxi \left[1 - \frac{L}{\sqrt{x^2 + L^2}} \right]$$

Show that the sum of the forces in the springs in the x-direction is given by

$$\vec{F} = -2kxi \left[1 - \frac{L}{\sqrt{x^2 + L^2}} \right]$$

$$F = 2i k (\sqrt{x^2 + L^2} - L) \left(\frac{x}{\sqrt{x^2 + L^2}} \right)$$

$$\Rightarrow -2kxi \left[\frac{\sqrt{x^2 + L^2}}{\sqrt{x^2 + L^2}} - \frac{L}{\sqrt{x^2 + L^2}} \right]$$

$$\Rightarrow -2kxi \left[1 - \frac{L}{\sqrt{x^2 + L^2}} \right]$$

(4)

- (vi) Toon aan (d.m.v. integrasie) dat die arbeid wat deur hierdie verskragte verrig word om die massa van $x = A$ na $x = 0$ te bewoog, gegee word deur:
Show (with integration) that the work done by these two spring forces to move the mass from $x = A$ to $x = 0$ is given by:

$$W = 2kL^2 + kA^2 - 2kL\sqrt{A^2 + L^2}$$

waar / where $\int \frac{x}{\sqrt{x^2 + L^2}} dx = \sqrt{x^2 + L^2}$

$$W = \int_A^0 F_x dx = \int_A^0 (-2kx + kL \frac{x}{\sqrt{x^2 + L^2}}) dx$$

$$W = -2k \int_A^0 x dx + kL \int_A^0 \frac{x}{\sqrt{x^2 + L^2}} dx$$

$$W = -2k \left[\frac{x^2}{2} \right]_A^0 + kL \left[\sqrt{x^2 + L^2} \right]_A^0$$

$$W = -2k \left(\frac{0^2}{2} - \frac{A^2}{2} \right) + kL (\sqrt{0^2 + L^2} - \sqrt{A^2 + L^2})$$

$$W = kA^2 - 2kL\sqrt{A^2 + L^2} + kL^2$$

$$W = 2kL^2 + kA^2 - 2kL\sqrt{A^2 + L^2}$$

if n sure

- (vii) Met $L = 1,2$ m, $k = 40$ N/m en $m = 0,5$ kg, bereken die energie wat in die vers gestoor is wanneer hulle 0,5 m uitgerek is.
With $L = 1,2$ m, $k = 40$ N/m and $m = 0,5$ kg determine the energy stored in the springs when they are stretched by 0,5 m.

$$U_x = 2kL^2 + kx^2 - 2kL\sqrt{x^2 + L^2}$$

$$U_x = 2(40)(1,2)^2 + (40)(0,5)^2 - 2(40)(1,2)\sqrt{0,5^2 + 1,2^2}$$

$$U_x = 0,4 \text{ J}$$

- (viii) Bepaal die spoed van die massa wanneer dit deur die ewewigsposisie gaan.
Determine the speed of the mass while it is moving through the equilibrium position.

Answer 0,323 m/s

Method Unknown

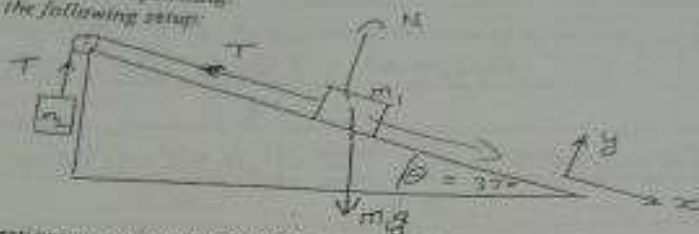
VRAAG / QUESTION 4

- (a) Kragte wat op 'n liggaam inwerk is sodanig dat $\sum \vec{F} \neq 0$. Gee Newton se tweede wet.
For different forces acting on an object $\sum \vec{F} \neq 0$. Give Newton's second law.

- (b) Gee 'n formule vir die statiese wrywingskrag F_s , indien die wrywingskoeffisiënt μ_s is.
Give a formula for the static frictional force F_s , with the static coefficient of friction μ_s .

$$F_s = \mu_s N.$$

- (c) Beskou die volgende opstelling:
Consider the following setup:



Indien die statiese wrywingskoeffisiënt $\mu_s = 0.4$ is moet ons die waardes van m_2 bepaal wat die sisteem in rus sal hou.

If the coefficient of static friction is $\mu_s = 0.4$ we have to determine the range of values for m_2 to keep the system at rest.

- (i) As m_2 klein is, sal m_1 neig om af te gly. / With m_2 small, m_1 would tend to slide down.

Die som van die kragte in y-rigting op m_1 is: / The sum of the forces in y-direction on m_1 is:

$$\sum F_y = N - m_1 g \cos \theta = 0$$

(1)

Die som van die kragte in x-rigting op m_1 is: / The sum of the forces in x-direction on m_1 is:

$$m_1 g \sin \theta - F_s - T = 0$$

(1)

- (ii) Spankrag T in die tou weens m_2 . / Tension in the rope due to m_2

$$T = m_2 (g + a)$$

- (iii) Die normaalkrag N op m_1 is: / The normal force N on m_1 is:

$$N = m_1 g \cos \theta.$$

(1)

- (iv) Met $m_1 = 10$ kg, toon aan dat $m_2 = 2.8$ kg moet wees. / With $m_1 = 10$ kg, show that $m_2 = 2.8$ kg.

$$m_2 = \frac{-\mu_s (m_1 g \cos \theta) + m_1 g \sin \theta}{g}$$

$$m_2 = \frac{-0.4 [(10)(9.8)(\cos 37^\circ) + (10)(9.8)(\sin 37^\circ)]}{9.8}$$

$$m_2 = \frac{27.67}{9.8} = 2.82 \text{ kg}$$

(4)

- (v) As m_2 groot is, sal m_1 neig om op te beweeg.
With m_2 large, m_1 would tend to move upwards.

Die som van die kragte in die x-rigting op m_1 (gee in simbole). / The sum of the forces in the x-direction on m_1 (give in symbols).

$$T - F_s - M_1 g \sin \theta = M_1 a$$

- (vi) Bepaal nou m_2 met $m_1 = 10 \text{ kg}$. / Now, determine the value of m_2 with $m_1 = 10 \text{ kg}$

$$M_2 = \frac{M_1 g \cos \theta}{g} + M_1 g \sin \theta$$

$$M_2 = \frac{0.4[(10)(9.8) \cos 37^\circ + (10)(9.8) \sin 37^\circ]}{9.8} = 9.2 \text{ kg}$$

- (vii) As $m_2 = m_1 = 10 \text{ kg}$, en $\mu_k = 0.3$ is, toon aan die versnelling van die sisteem is $a = 0.78 \text{ m/s}^2$.
With $m_2 = m_1 = 10 \text{ kg}$, and $\mu_k = 0.3$, show that the acceleration of the system is $a = 0.78 \text{ m/s}^2$.

Take the incline a positive

$$a = \frac{M_2 g - M_1 g \sin \theta - F_k}{M_1 + M_2}$$

$$a = \frac{(9.8)(10) - (10)(9.8) \sin 37^\circ - 0.3[(10)(9.8) \cos 37^\circ]}{20}$$

$$a = \frac{98 - 58.77 - 23.48}{20}$$

$$a = 0.77 \approx 0.78 \text{ m/s}^2$$

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VRAAG / QUESTION 5

- (a) Definieer hoeksnelheid ω in gte sy eenhede.
Define angular velocity ω and give it's units.

Rate at which angular position of object changed with time $\text{rad/s}^\wedge$ (2)

- (b) Toon aan dat die verband tussen lineêre snelheid v en hoeksnelheid ω gegee word deur $v = \omega r$.
Show that the relation between linear velocity v and angular velocity ω is given by $v = \omega r$.

$$v = \frac{ds}{dt} = r \frac{d\theta}{dt}$$

$$v = r \omega$$

- (c) 'n Voorwerp wat met hoeksnelheid ω roteer word opgedeel in deeltjies Δm_i wat 'n lineêre snelheid $\vec{v}_i$ het. Δm_i is op afstand r_i vanaf die rotasie-as. Bepaal die rotasie kinetiese energie en toon aan dat die traagheidsmoment $I = \int r^2 dm$.

An object rotates with angular velocity ω and is divided in small particles Δm_i , each with linear velocity $\vec{v}_i$. Δm_i is at a distance r_i from the axis of rotation. Determine the rotational kinetic energy and show that the moment of inertia is $I = \int r^2 dm$.

$$K_i = \frac{1}{2} m_i v_i^2$$

$$K_R = \sum_i K_i = \sum_i \frac{1}{2} m_i v_i^2 = \frac{1}{2} \sum_i m_i r_i^2 \omega^2$$

$$\Rightarrow I = \sum_i m_i r_i^2$$

$$K_R = \frac{1}{2} (\sum_i m_i r_i^2) \omega^2$$

$$K_R = \frac{1}{2} I \omega^2$$

$$\int I = \int r^2 dm$$

- (d) 'n Gedeelte van 'n dun uniforme staaf het 'n massa dm en sy lengte is dl . Skryf 'n uitdrukking neer vir die lyndigtheid λ van dl . Gee ook die eenheid van λ .

A part of a uniform thin rod has a mass dm and its length is dl . Give an expression for the line density λ of dl . Also give the unit for λ .

$$dm = \lambda dl$$

$$\lambda = \frac{dm}{dl} \quad \text{M/L}$$

- (e) Beskou 'n dun uniforme staaf met massa M en lengte L . Toon aan die traagheidsmoment om 'n as loodreg op die staaf en deur sy swaartepunt gelyk word deur $I = \frac{1}{12} ML^2$.
A thin uniform rod has a mass M and length L . Show that the moment of inertia about an axis which is perpendicular to the rod and which goes through its center of mass is given by $I = \frac{1}{12} ML^2$.

$$\begin{aligned}
 dm &= \lambda dx = \frac{m}{L} dx \\
 I_y &= \int r^2 dm = \int_{-L/2}^{L/2} x^2 \frac{m}{L} dx \\
 &\Rightarrow \frac{m}{L} \int_{-L/2}^{L/2} x^2 dx \\
 &\Rightarrow \frac{m}{L} \left[\frac{x^3}{3} \right]_{-L/2}^{L/2} = \frac{m}{L} \left[\frac{(L/2)^3}{3} - \frac{(-L/2)^3}{3} \right] \\
 &= \frac{1}{12} ML^2
 \end{aligned}$$

(5)

- (f) Wanneer die rotasie-as steeds loodreg op bogenoemde staaf is, maar deur die staaf se een eindpunt gaan, bepaal die staaf se traagheidsmoment om hierdie as deur gebruik te maak van die teorie van parallelle asse.
With the axis of rotation is still perpendicular to the rod mentioned above, but it goes through its one end, determine the moment of inertia about this axis by applying the parallel-axis theorem.

$$I = I_{cm} + Md^2$$

$$= \frac{1}{12} mL^2 + M \left(\frac{L}{2} \right)^2 = \frac{1}{3} ML^2$$

(3)

- (g) Elkeen van die drie lemme van 'n helikopter se rotor het 'n massa van 240 kg en is 5,2 m lank. Toon aan dat die traagheidsmoment van die rotor is $I = 6,49 \times 10^3 \text{ kg m}^2$.
Each of the three helicopter rotor blades is 5,2 m long and has a mass of 240 kg. Show that the rotational inertia of the rotor is $I = 6,49 \times 10^3 \text{ kg m}^2$.